Function Notation Is Not Multiplication

Directions

- Every problem below shows work that treats $f(x)$ as "f times x". Your job is to repair the computation *and* to say what the notation really asks for.
- $f(3)$ means: put 3 in for the input everywhere it appears in the rule, then simplify. The letter naming the function is never a factor.
- Write the corrected value on the blank, and the diagnosis on the ruled space.

1. Let $f(x) = 4x - 7$. A student wrote

$$f(3) = f \cdot 3 = 3f.$$

(a) Evaluate $f(3)$ correctly. $f(3) =$ _________

(b) In one sentence, say what the notation $f(3)$ actually instructs you to do.

2. Let $g(x) = x^2 - 2x$. A student wrote

$$g(5) = g \cdot 5^2 - 2 \cdot g \cdot 5 = 25g - 10g = 15g.$$

(a) Evaluate $g(5)$ correctly. $g(5) =$ _________

(b) Name the error in the student's first step.

3. Let $f(x) = x^2{+}1$. A student claims that $f(2{+}3) = f(2){+}f(3)$, "because the parentheses just multiply".

(a) Compute both sides. $f(5) =$ ________ and $f(2) + f(3) =$ ________

(b) Explain why the claim fails for this function.

4. Let $f(x) = 6x - 4$. Asked to solve $f(x) = 20$, a student wrote

$$f \cdot x = 20, \qquad \text{so} \qquad x = \frac{20}{6} = \frac{10}{3}.$$

(a) Solve $f(x) = 20$ correctly. $x =$ ________

(b) Name the error in the student's first line.

5. A print shop charges a one-time setup fee plus a fixed price for each shirt. The cost of an order of n shirts, in dollars, is $C(n) = 12n + 25$. A student found the cost of an order of 4 shirts by writing $C \cdot 4 = 4C$.

(a) Find $C(4)$, the cost of that order in dollars. $C(4) =$ ________

(b) Explain what the 25 represents, and why $C(4)$ is not "4 times C".

6. A tank holds 45 litres and drains at 5 litres per minute, so the amount left after t minutes is $A(t) = 45 - 5t$. Asked when the tank is empty, a student wrote "A times t is 0, so $t = 0$".

(a) Solve $A(t) = 0$ correctly, in minutes.

Answer: ________________ min

(b) Name the error in the student's reasoning.

7. Let $f(x) = 4x - 12$. The two questions below look almost alike and are not the same question.

(a) Find $f(0)$. $f(0) =$ _________

(b) Solve $f(x) = 0$. $x =$ _________

(c) In one sentence, explain the difference between what (a) asks for and what (b) asks for.

8. Let $f(x) = x^2 + 2x$. A student argues: "$f(x)$ means f times x, so doubling the input doubles the output — that is, $f(6) = 2f(3)$."

(a) Compute both values. $f(3) =$ _________ and $f(6) =$ _________

(b) Show that $f(6) \neq 2f(3)$, and explain what $f(6)$ really instructs you to do.